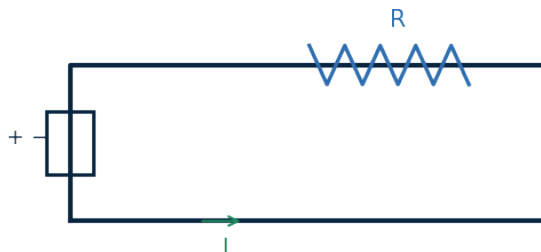


Current Electricity

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Electric current $I = Q/t$ (charge flowing per unit time). Current density $J = I/A$.
- Ohm's Law: $V = IR$, valid for ohmic conductors at constant temperature. Resistance $R = \rho L/A$, where ρ is resistivity.
- Resistors in series: $R_{eq} = R_1 + R_2 + \dots$ (same current through each). Resistors in parallel: $1/R_{eq} = 1/R_1 + 1/R_2 + \dots$ (same voltage across each).
- Kirchhoff's Current Law (junction rule): the sum of currents entering a junction equals the sum leaving it (conservation of charge).
- Kirchhoff's Voltage Law (loop rule): the algebraic sum of potential differences around any closed loop is zero (conservation of energy).
- Electrical power $P = VI = I^2 R = V^2/R$. Heat generated in a resistor: $H = I^2 R t$ (Joule heating).
- EMF of a cell vs terminal voltage: $V = E - Ir$, where r is internal resistance and I is the current drawn.
- Wheatstone bridge is a balanced circuit used to accurately measure an unknown resistance by comparison.



A simple circuit: battery, resistor R , and current I flowing around the loop.

2. Formula Sheet

Ohm's law	$V = IR$
Resistance from resistivity	$R = \rho L / A$
Series combination	$R(\text{eq}) = R_1 + R_2 + \dots$
Parallel combination	$1/R(\text{eq}) = 1/R_1 + 1/R_2 + \dots$
Electrical power	$P = VI = I^2R = V^2/R$
Terminal voltage of a cell	$V = E - Ir$
Heat generated (Joule's law)	$H = I^2 R t$
Wheatstone bridge balance condition	$P/Q = R/S$

3. Practice Problems (22 questions)

1. A current of 2 A flows through a resistor of 10 ohm. Find the potential difference across it.
2. Find the resistance of a wire of length 5 m, area of cross-section 1 mm^2 , and resistivity $1.7 \times 10^{-8} \text{ ohm m}$.
3. Three resistors of 2 ohm, 3 ohm, and 5 ohm are connected in series. Find the equivalent resistance.
4. Three resistors of 2 ohm, 4 ohm, and 4 ohm are connected in parallel. Find the equivalent resistance.
5. A cell of EMF 6 V and internal resistance 0.5 ohm is connected to an external resistance of 5.5 ohm. Find the current drawn from the cell.
6. Find the power dissipated in a resistor of 20 ohm carrying a current of 3 A.
7. A wire carries a current of 5 A for 10 minutes. Find the total charge that flows through it.
8. State Kirchhoff's junction rule and loop rule, and explain the conservation principle behind each.
9. An electric heater rated 1000 W operates at 220 V. Find its resistance and the current it draws.
10. Two resistors, 4 ohm and 6 ohm, are connected in parallel, and this combination is connected in series with a 2 ohm resistor. Find the total resistance of the circuit.
11. A copper wire and an aluminum wire have the same length and cross-sectional area. Given resistivity of copper is $1.7 \times 10^{-8} \text{ ohm m}$ and aluminum is $2.7 \times 10^{-8} \text{ ohm m}$, find the ratio of their resistances.
12. Explain the working principle of a Wheatstone bridge and state its balance condition.
13. A battery of EMF 12 V and internal resistance 1 ohm is short-circuited. Find the current that flows.
14. Find the energy consumed (in kWh) by a 100 W bulb operating for 5 hours.
15. Explain why the resistance of a metallic conductor increases with an increase in temperature.
16. A current of 0.5 A flows through a 100 ohm resistor for 2 minutes. Find the heat generated.
17. Two cells of EMF 2 V and 4 V with negligible internal resistance are connected with their positive terminals together. Explain qualitatively what happens (conceptually, no calculation needed).

18. Find the drift velocity of electrons in a copper wire of cross-sectional area 1 mm^2 carrying a current of 1 A ($n = 8.5 \times 10^{28} / \text{m}^3$, $e = 1.6 \times 10^{-19} \text{ C}$).
19. State the difference between EMF and terminal voltage of a cell.
20. A wire of resistance R is cut into 4 equal parts, and all 4 parts are connected in parallel. Find the new equivalent resistance in terms of R .
21. Explain why household electrical appliances are connected in parallel rather than in series.
22. A 60 W bulb and a 100 W bulb are both rated for 220 V . Find the ratio of their resistances.

Answer Key

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|---|---|
| 1. $V = 20 \text{ V}$ | 2. $R = 0.085 \text{ ohm}$ |
| 3. $R(\text{eq}) = 10 \text{ ohm}$ | 4. $R(\text{eq}) = 1 \text{ ohm}$ |
| 5. $I = 1 \text{ A}$ | 6. $P = 180 \text{ W}$ |
| 7. $Q = 3000 \text{ C}$ | 8. Conceptual - junction rule from charge conservation;
loop rule from energy conservation |
| 9. $R = 48.4 \text{ ohm}$, $I \text{ approx } 4.55 \text{ A}$ | 10. $R(\text{total}) = 4.4 \text{ ohm}$ |
| 11. Ratio (Cu:Al) = $1.7 : 2.7$ (approx $0.63 : 1$) | 12. Conceptual - balanced bridge has no current
through the galvanometer when $P/Q = R/S$ |
| 13. $I = 12 \text{ A}$ | 14. Energy = 0.5 kWh |
| 15. Conceptual - increased thermal vibration of ions
scatters more electrons, increasing resistance | 16. $H = 600 \text{ J}$ |
| 17. Conceptual - current flows driven by the higher
EMF cell, direction depends on net configuration | 18. $v_d \text{ approx } 7.35 \times 10^{-5} \text{ m/s}$ |
| 19. Conceptual - EMF is the total energy per charge
from the source; terminal voltage is EMF minus the
drop across internal resistance ($V = E - Ir$) | 20. $R(\text{new}) = R/16$ |
| 21. Conceptual - parallel connection keeps voltage
same across each appliance and allows independent
operation | 22. Ratio ($60\text{W}:100\text{W}$) = $5:3$ (R inversely proportional
to power) |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.